Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/05/2019
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight,Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/22/2016	**End Date:**	06/21/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	January 2018 report: Prof. Alain Karma is no longer a Co-Investigator on this project, but continues to be a collaborator, who is responsible for the modeling part of the project through a separate NASA grant.		
COI Name (Institution):			
Grant/Contract No.:	NNX16AL77G		
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Performance Goal Text:			
	NOTE: Continuation of "Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)" grant number NNX12AK54G, with the same Principal Investigator Dr. Rohit Trivedi. During the manufacturing of industrial components by solidification processes, which includes casting and welding, the properties and the reliability of the finished products depend on the resulting microstructure that is governed by the processing conditions imposed during the solidification process. It is thus crucial to quantitatively relate the detailed microstructural features with the growth conditions and the composition of the alloy so that one can design alloy composition and growth conditions to optimize processing routes that will produce the object with superior properties. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced		

Task Description:	melt convection under terrestrial conditions. Extensive experimental and theoretical studies over the past several decades have yielded basic insights into the formation of complex solidification patterns; their dynamics in three dimensions (3D) still remain poorly understood since fluid convection effects on Earth lead to large scale variation in temperature and alloy concentration in bulk samples that are quite complex to quantify. The presence of these convection effects make it difficult to probe the spatially extended pattern dynamics under well-controlled diffusive growth conditions with imposed constant growth velocity, thermal profile, and alloy composition. To minimize convection effects, detailed experimental studies of interface dynamics on Earth have been restricted to thin samples where convection effects are minimized. However, in thin samples it is not possible to study the formation of extended three-dimensional patterns. Also, the dynamics of microstructure evolution is strongly influenced by the thickness of the sample. Therefore, directional solidification experiments are designed on International Space Station (ISS) where convection effects are negligible so that precise evolution of three-dimensional microstructure can be investigated and characterized quantitatively. These results can then be used to develop appropriate theoretical models and to validate these models quantitatively to confirm the predictive nature of the models. Under conditions of very low velocity and low composition, with high thermal gradient, solidification occurs with a planar interface. The most common solid-liquid interface patterns that form during the solidification in metals, however, consist of complex spatially extended arrays of cells or dendrites that form when the growth conditions exceed the critical condition for the onset of morphological instability of a planar interface. In previous microgravity experiments, Directional Solidification Insert (DSI) experiments, we examined the time evolution of the three-dimensional patterns under a wide range of velocity for two different thermal gradients in succinonitrile-0.24 wt% camphor. For this dilute composition, the microstructures were mainly planar and cellular. Dendritic arrays were observed at higher velocities, although at higher velocities significant latent heat effects caused the macroscopic interface to become curved so that only very limited velocities could be examined in the dendritic regime. In the current study, DSI-R experiments, carried out on the International Space Station, a higher alloy composition, succinonitrile-0.5 wt % camphor, was used to examine the dynamics of 3D dendritic arrays at lower velocities. In addition, the effect of composition on the spatiotemporal evolution of three-dimensional cellular and dendritic array structures will also be characterized. Since one of the goals is to examine the formation of dendritic arrays that are characterized by the presence of side branches, specific emphasis was placed on the fundamental understanding of the origin of side branch formation and the subsequent dynamics of side branch propagation in the two-phase region or the mushy zone, between the primary dendrite arms. These sidebranch patterns are directly related to the scale of microsegregation that controls the properties and reliability of cast objects. Detailed experimental studies were carried out in a microgravity environment where convection is reduced sufficiently to obtain growth under diffusive conditions. The interface evolution in three dimensions was directly visualized through two cameras that viewed the interface from the side and from the top. This continuous observation of the interface provided an unprecedented opportunity to elucidate fundamental mechanisms of the formation of those hierarchical structures by providing quantitative benchmark data that can be compared to predictions of computational models. The goal of this work is therefore to (i) obtain extensive benchmark experimental data under microgravity conditions over a range of velocity, thermal gradient, and alloy composition in a transparent system by imaging the dynamic evolution of microstructures under directional solidification conditions in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) installed onboard the ISS, and (ii) use state-of-the-art phase-field simulations to make quantitative predictions of dendritic hierarchical structure formation under diffusive growth conditions that can be directly compared to experimental observations under prescribed growth conditions. We expect the combination of those experimental and computational studies to establish a reliable framework for predicting dendritic microstructures in a wide range of alloy systems of technological relevance. This research project is carried out in collaboration between NASA and CNES (Centre national d'études spatiales (CNES), the French government space agency) and includes experiments on International Station, experiments under terrestrial conditions in thin and bulk samples, and the development of rigorous theoretical models to compare with the experimental data obtained terrestrially in thin samples and from International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based super alloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process that is used to fabricate appropriate components. These microstructures in turn control the integrity and mechanical properties of cast parts. By combining well-controlled and accurately characterized experimental evolution of three-dimensional solidification microstructures with quantitative phase-field simulations, this research should bring significant progress in understanding the origin and propagation of side-branching and the nature of the cell to dendrite transition, which have been a long standing issue in solidification for several decades. The deeper understanding of microstructure formation dynamics, including the development of side-branched structure in the mushy zone, shall thus accelerate and optimize the design of new technological alloys and their processing conditions that should directly benefit US industries.
Task Progress:	In the current set of experiments on the International Space Station (ISS), in DSI-R, we carried our analogous experiments in succinonitrile-0.5 wt % camphor to examine the effect of composition on the formation and spatiotemporal evolution of three-dimensional cellular and dendritic array structures. In addition, specific emphasis will be placed on the fundamental aspects of the origin of side branch formation and the dynamics of side branch propagation in the two-phase, or the mushy zone, between the primary dendrite arms. During this research period, we carried out the following tasks: (I) Experimental studies on International Space Station, (II) developed a rigorous theoretical model of primary spacing model that included the complex thermal profile in the adiabatic zone, and (III) ground-based studies in thin samples to develop programs to input periodically varying sample motion and carry out experiments to correlate the oscillations in input velocity with the development of side arms. In summary, in this research grant period we have carried out ground-based experiments in thin sample to study the effect of imposed oscillatory velocity on dendritic microstructure with an emphasis on the effect of initial velocity, amplitude of oscillations, and the frequency of oscillations on side branch formation. We have developed software to programs for Compumotor to produce sinusoidal velocity profile for the sample translation. Preliminary sets of experiments have given us a wealth of information on the effect of imposed oscillations on dendritic microstructure. They have shown that oscillation influences the side branch formation for lower velocities when the characteristic time is large. The next step in this research is to characterize the details of the effect of oscillation on the location of side branch

	formation under different frequencies and under higher thermal gradient values. These results also show that experiments at lower velocities, i.e., 1.5 and 2.0 µm/s with large characteristic time should be used in experiments on ISS. More studies will be carried out in this direction to establish the final experimental matrix for use on ISS experiments.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Song Y, Tourret D, Mota FL, Pereda J, Billia B, Bereon N, Trivedi R, Karma A. "Thermal-field effects on interface dynamics and microstructure selection during alloy directional solidification." Acta Materialia. 2018 May 15;150:139-52. https://doi.org/10.1016/j.actamat.2018.03.012 , May-2018